

ActInf Livestream #025.1 ~ "The Computational Boundary of a Self"

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hello everyone welcome to acton flab live stream number 25.1 today is july 6th 2021 and we're going to be talking about the paper the computational boundary of a self developmental bioelectricity drives multicellularity and scale-free cognition we are a participatory and online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is a recorded and archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here we will be following good video etiquette for live streams here at the short link you will see all of the live streams and different series that we've done leading up to this in the communications unit of the active inference lab and today we're going to be jumping into the first half of a discussion of the paper uh in july and on the 13th we will have 25.2 which is also on this paper and hopefully with the author joining today in act in flab live stream number 25.1 we're going to be learning and discussing this paper uh so i will um pass it off i'm blue and i am an independent research consultant based out of new mexico and i will pass it to steven for a quick round of introductions thanks blue hi so this is stephen here i'm based in toronto um i'm doing a practice-based phd and uh i'm interested in how active inference can help unify lots of threads that have been very confusing to me over the last few years and if it's okay i'm gonna throw this over to dave douglas yes hi i'm in the philippines up in the mountains uh the rainy season has begun and we just survived the fourth of july so all is good thanks back to dan i guess thanks uh i'm daniel i'm in davis and yep looking forward to this point one discussion and also nice job with the broadcasting and facilitating blue thanks so we can just start off with just some questions about what you liked about the paper or what was something you remembered or something that you're curious about well i am embarrassed to say i didn't read it at all and i was you know i i started looking over it as oh my goodness we're going to talk about wilhelm reich we're going to talk about orgones and orgasms and cloud busters and and living flying saucers and i guess we're not going to get any of that so i don't

know cool well the aims and claims of the paper um so here the paper the author says the aims were to define individuals themselves in a way that facilitates taxonomy comparison and communication with evolved created biological artificial and exobiological agents and he proposes a fundamental definition of an individual based on the ability to pursue goals at an appropriate level of scale and organization and suggests a formalism for defining and comparing the cognitive capacities of highly diverse types of agents what does anybody think about that yeah i i think that the multi-scale part of this is very very interesting very exciting because this is one of the big offerings that active influence has and it's been offered in a lot of simulations in silico context but to have it actually described from a real biological kind of cellular perspective and is for me really really useful because i'm kind of looking partly at how the heuristics of all this are as much as the you know the mathematics of it so yeah i think that's really really interesting cool daniel i liked the part about taxonomy because that kind of speaks to this evolutionary worldview and it always seemed interesting to me studying collective behavior like in the ants that there were really precise ways of talking about comparing some physical properties in biosystems like how hard is the material or its density and then there was less of a way to talk about similarities in cognitive processes whether on an organism or how they interacted with their niche especially so just the idea that there could be a new sort of tool in the toolbox to help us compare different cognitive systems that might have hybrid aspects like computer and human or agent and niche those are kind of cool areas to explore so i think that's a big topic definitely like the other maybe um like comparison of like cognitive capacities isn't really related to cognition but it's integrated information theory right so you know when you can compare like the consciousness of a slime mold versus the consciousness of a human this kind of stands uh alongside of that in in terms of you know being able to to maybe not quantitatively yet but hopefully um you know people will run with this and be able to apply some quantitative methods to that uh cool so he the author also proposes a plausible naturalistic framework for the evolutionary scale-up of cognition from the earliest origins of life and hypothesizes about the forces that drove it and the major transitions along the continuum and the goal is to show how complex agency and goal directedness evolves naturally from ancient mechanisms uh so some questions how do multiple nested scales of individuality work uh that's super interesting to me let's see um daniel yeah well i wrote the question so maybe i'll ask or bring up why i thought it was interesting uh is there like a total value of cognition in the full full stack here the full cone and then there's going to be certain levels that are more relevant than others because they're a total amount of

individuality or are there just certain ways that like the the cones for example don't overlap like there's certain spaces almost that are implied that are more than the sum of the parts and so how's how are we gonna even just first draw it out think about it actually one thing that comes to mind when you talk about the sum of the parts i think what i i'm interested by when he talks about a cell and then he talks about these groups of cells and how they are sort of joined and there's this kind of bioelectric dynamic that sort of helps them communicate i think that's a really um making a bit of a jump here but i think it's interesting to the difference between people communicating in a community context where there's this kind of resonance going on between people and what it's like when there's just separate entities sending messages to each other so to speak um there's something about this um ability to form some sort of bioelectric um dynamic which i think is in itself quite interesting as a heuristic yeah for sure daniel agreed we're gonna bring it go to bioelectricity soon because i think with um the author visiting next week it'll be good to get some of these notions that we're having about where bioelectricity plays a role like those are some of the big theoretical questions like what is individuality across levels is it informational is it evolutionary is it functional all these questions about like holism and reductionism more than the sum of the parts what is collective behavior what is emergence there's that whole body of thinking and then the piece that's tying it all together is the bioelectric and there's a few reasons why in the paper that is being proposed as important like the primacy or the primitiveness um and also just the um signaling dynamics and the what it appears to allow cells to be able to do we can explore that later so agree like how is bioelectricity supporting this broader story about multi-scale systems which doesn't just just by drawing it out this way doesn't imply bioelectricity so where is bioelectricity coming into the picture we'll keep thinking about that yeah absolutely i'll be we'll talk about that later i think and it's it's something that i don't think and correct me if i'm wrong other papers on active inference they have not tended to come at that aspect it's it's they've tended to come at it from a in silico kind of statistical state-based process so this this does give something that's a bit more um organic i suppose potentially so i think that there's maybe there's much more that can be drawn from that going forward so thanks yeah i think the um you know people have done more organic uh stuff but it's always like even related to bioelectricity but it's always related to neural bioelectricity as opposed to non-neural bioelectricity which is what kind of drives this paper forward and the you know of course there's a relationship between bioelectricity and the homeostatic drive which the author talks about as driving the major evolutionary transitions and you know probably much of what we

do much of how organisms think and behave is driven by you know that that drive for uh homeostasis so uh more questions what do we learn or enable when we understand multi-scale individuality daniel maybe one way to frame what we could learn or enable here would be like in a read and write with respect to designing new kinds of multi-skill systems that's like the right and then to look around and discover the kinds of multi-scale systems that we're already interested in with like reading so it's kind of there's an interpretive ability when we can look across multi-skill systems but then also within a virtual machine or within some sort of physical system there's ways to just design systems that have certain kinds of relationships between the parts and the holes like a governance system where such and such a distribution of of uh influence exists or something like that stephen stephen one area that's been of interest and i think this can relate to is in an area called mental space psychology they've been looking at peripersonal neurons the mirror neurons as people talk about them as well and how they resonate and i think that um this that ability to get that dynamic relationship between people or with objects in our shared model and the ability for these to see that there's some sort of resonance in terms of how things relate to others in space that that seems to tie into what you were saying about the bioelectrics of um neurons tightenings have been sort of related to this and ties into that ability for maybe us to re or reconnect some of um inference processes across distance you know because obviously the cells they're touching each other there's a plausible root but how do we maybe get some of these resonances between cells which are not touching each other or even organisms which are not touching each other or maybe they need to touch and that gets us into intimacy and stuff but yeah good point that like brings us so i just pulled up the slide um talking about gap junctions which these are like you know it's their little holes that enable you know those proximal cells to share information essentially um about their environment or their intracellular um like state of existence and that's a cool um way for you know that it facilitates the the informational spread um and we see that in the figure also let's see let's come to that figure um so this one yeah so that gives you like you know if if there was a boundary like a solid boundary or maybe even a lipid bilayer that wasn't didn't have a little hole in it connecting cells then there would just be the individual spatial perception that's shown here in figure three but because of the gap junctions um you know a cell communicates to all the proximal cells and this expands the spatial perception and the memory and the anticipation and so it's like this group integration here uh dave yeah going even farther into field effects um i heard a presentation by some students in china recently who were tracking down non-uh tracking down electrical effects in mammalian nervous systems that didn't that

were not confined to local chemical and electrical effects so they didn't even need to use the pores they said this is not dominant but it's there and under certain circumstance it does become pretty important like in um long-term potentiation they're saying that's part of what's going on that you get these standing waves and they're then they're uh conjecturing that part of the classic brain waves are literally are waves in the brain rather than a summation of these local uh synaptic jumps don't know daniel oh daniel yeah just one comment on what dave just said there about these kind of traveling fields that is really related to kristin and collaborators work with spm which was basically a toolkit for looking at dynamical systems and then integrating imaging based modalities for like neuroimaging and with the field-based measurements like eeg so kind of taking a certain anatomical shape into account when you are integrating these different projections or different modeling ways of kind of two different uh perspectives on the same underlying system so that's kind of an interesting link that you just made there yeah it's hard to under it it's hard to overestimate the importance of professor fristen's enormous amount of hands-on work with interpreting brain direct physical brain data and how important that is in just implicitly shaping how he uh approaches uh these system activities and system effects yeah i keep yeah i keep slapping myself wait a minute what does this guy spend the last 40 years doing you know don't get stuck in the peculiarities of the mathematical descriptions he has a real reason for doing this stuff that way dania you go first stephen oh okay yeah sure well i suppose being a bit speculative here but with the brain waves in the way that we've got this bioelectrics on the cell i mean different scales of i think this is an open question but it certainly seems to have plausible roots to being thought about is how much of these brain waves or this electrical activity across regions of the brain analogous or in some way similar to the the descriptions at the cellular and multi-cellular level so i think that's a good point that dave brings up there thanks daniel yep great point stephen and that's one of the main i think arguments or the ways that the field or understanding of biology would sort of shift if bioelectric were given a larger role there'd be more of a continuum of chemical and electrical function between internal nervous systems and the epithelial cells which like we talked about in the dot zero are closely related through development and other cell types and seeing that as more of a unifying framework within an organism's cognition as well as in an evolutionary sense like all these different affordances that the bioelectric reconfiguration allows for which is some of the work that's cited in the paper and has continued since then is just empirical demonstrations that the bioelectric field can like lead to different morphologies and so then there's kind of a way to move on that phenotype landscape that doesn't involve like

only the um sequence based remodeling of gene regulatory networks like transcription factor binding site evolution could still be important and also there could be certain points where the bioelectricity enables new uh things to new phenotypes to change as well yeah definitely and um maybe a little bit in this paper but a lot of mike's work is really great uh talking about um or or demonstrating you know the the role of bioelectricity in shape and you know when you think about like you there the form and the function are just coupled together so like the function of a cell and the shape of a cell and all that is is really um interesting and how bioelectricity is connected to the form and the um function uh stephen yeah and i think this also ties or certain gives reason to look at some other people who presented such as the work on quantum contextuality that we had and that some of the information geometry work and there's the new paper on um bayesian mechanics and active inference this whole idea of the shape of information which i mean that is very cutting edge but i think they they don't seem incompatible let's put it like that i don't want to put my name to how they're compatible exactly but this certainly seems to be an awful lot of things here that would suggest their plausible connection definitely uh daniel oh daniel so one other aspect there's the extended spatial that's this horizontal representation where whether it's the bacteria that are aggregating in the biofilm or the development of the embryo or whatever extended spatial system maybe even like networks of computers or something like that that's extended spatial extent in the snapshot and then there's also blue going down visualization here like with memory fading off into the past and then also the green like the anticipation so that's the current state anticipating out forward so just kind of using a uh visualization of kind of space going out and time as this axis that goes through the current moment that turns out to be something that visualizes the past present and future really nicely but then also like we saw with the light cones and probably other approaches it kind of sets up the integration of memory action in the moment and anticipation in a way that's almost like set up to be addressed with a commonality rather than a future oriented framework that has to bolt on this like memory module it doesn't really make sense but this is a more integrated way to think about the past present and future nice uh stephen yeah i like thanks daniel for bringing that up actually that's really helpful and and i one thing that i find curious as well is even though we're talking about cells we think about how that diagram puts the future up and the past down right as we think about could have done it the other way but that's naturally how we look up and the air you know to see things and things go down and a bit more in the earths and that's where our history is so even though it probably was not conscious that would have just felt like the right way to design the diagram

you see how um you know the way we make our meaning is embodied as well the way we make our diagrams is in ways embodied yeah nice definitely daniel so that steven that kind of uh whether something's pointing up or down it reminds me of in evolutionary biology where some subfields would have a tree that branched up so it looked like a tree other fields would show it branching down and then there was like apparently disciplinary like debates over which which rotation of the same information it quote really was is it a tree or is it you know something else but then that was related to a question that i just had about the visualization so it's cool that these panels were kind maybe something we can ask about next week they seem to be made like in collaboration with an artist so that sounded like something kind of interesting to have in a scientific paper it helps it be more discussable sometimes but maybe there's a learning experience for the people involved and then also just like it's pretty easy to visualize the chemical part like we have these beautiful you know walking protein visualizations kind of like okay proteins kind of look like that and they bump into each other but then how will we visualize the role of the bioelectric greeting scott nice to have you how will we visualize that um in new ways maybe by integrating the future and the past into it or i don't know maybe there's some other way like the diamonds that we saw so yeah yeah i think that's that's to be a really good question to ask actually because um yeah i work quite a lot with in graphics studios where they do different types of diagrams for different things and i think there's a lot of stuff that conventions that are out there so they must have had also bring in and sort of make it make sense in the scientific paradigm but i do like the way that they've they've used the term perception there and brought that in for a cell i think that's um again unusual but makes sense if we think of perception in a much more deflationary sense than how people traditionally think about perception and active inference as well as just some sort of way of knowing um so i really and again that also starts to bring in like cells when we think about what is the meaning that's not how a cell is cells could have a sense of knowing and there could be a hundred billion cells knowing which is pretty much what indigenous knowledge says about knowing so it's another it ties in with that regime of knowing which is often minimalized and excluded so i think this is quite quite useful nice and just kind of to talk about like different renditions of like the light cone i mean this is a question that i definitely have for the author uh here this is like not the traditional way that i'm used to seeing a light cone so this in this slide you can kind of see how it's more of like a conical shape as opposed to like a diamond um and like in space and time um and even like so this is up and down um but you know there's also this rendition that i've seen as well that's left and right so like from the from

the state of right now to the future and so it's interesting to think about why is the space and time bigger in the present and going to a more narrow point in the future like for me that's kind of not how i'm used to conceptualizing um these diagrams so i think about the space of right now like i can't do anything other than what i'm doing right at this moment but you know potentially the the as time goes on my my state space gets bigger if that makes sense potential state space i i really like that as well because the thing is it's interesting that cone is a bit it's very popular in this strategic foresight field of looking at futures and strategic futures work but there is a slight like when it's done as a cone going into the future it's very visual it's like it's like using the analogy of a beam of light effectively again using one modality and it assumes it's kind of travels in a straight line and it projects and it's whereas when you start to get into this non-linear swarming type thing i mean in some ways it's you just see what rises up in the future you know i mean it's like what what structures emerge into the future um they don't project into the future in the way that a lot of strategic foresight so i think this does that question about how those two different models manifest and get used and how this other type of model actually while it's not as easy to intuit may actually be more more i don't know if we say accurate but yeah maybe more accurate is better way of saying it yeah i i actually um definitely agree because you know not to be morbid but we all are progressing to the same end right so and as we age our affordances are less and less like you know we may we might not be able to you know cognate as well or or we might be less mobile as we as we get older uh daniel sorry did i take the words out of your mouth um how about dave first and then i'll go okay dave yeah so is that becoming more focused into the future does that have anything to do with most of us becoming more and more subject matter experts rather than the other kind of expert daniel nice question dave so um the the sort of duality of the time cone and the time diamond it just it just suggests that there's certain things that we want to think about as being maximized in the present maybe and certain things that are being as if they're minimized in the presence so it's just two different and we'll figure out maybe in some ways there's some things that we know off into the future and there's other things that are maximally constrained in the present but then to connect that to the stephen's question about um strategy and um you know ask scott what he was uh interested in with uh you know joining us it made me think about the width of this cone going out into the future or the the length of the one that's more like a it's kind of like the horizon which you need to consider for de-risking or for strategy and so that space just like the butterflies wings that space expands super rapidly now you might have a hypothesis about the space that only one region is likely to be um explored or that all of your

strategy would would you know work across this whole space but it just sets up the discussion instead of like time points and bifurcating points it's like actually it's going out really vastly so there has to be some more adaptive way of realigning because the time cone is going to be too large to search through in the future even on a short time step so how could we enact a strategy in the current given what something what we are maximizing so that we could actually have an adaptive strategy moving forward nice dave and then stephen no i just hit the button wrong got it stephen yeah i saw scott's got his ice cream cones there so i'm gonna be curious to hear his his point but um tying with what we're saying is also we have this because swarming dynamics and non-linear dynamics basically aren't part of the most systems approaches really there's an assumption that it's a volume of information problem going from the present to the future and it's just finding ourselves in all of but um and at times when we've got the like in modern world we've kind of thought the world is much more predictable than it is and maybe it was when it was in a very constrained system but now you know we can see that even in our own social systems the things can once they bifurcate that whole cone thing breaks down because it suddenly it's it just um so i think that there's there's there's something quite um useful here actually in rethinking and using this alternative or maybe something which is a other version of this um this this this version of this cone um which doesn't rely on ocular light visual because there's no eyes in for instance they're not seen into the future right there that's only something that happens at a quite a aggregated organism so um this is this is actually probably has some other spin-offs that's quite useful scott this is a fascinating conversation so by the way this is a cone that i saw in new york that it was an ice cream cone it was very impressive it had nine scoops on it so that was that's the kind of light cone i like the um you know i was thinking back of that now the knowing that you were saying before blue and it's so interesting so what if we perceived a group as a bunch of cones a bunch of simultaneous cones and because if they have embodiments or shared meaning physically or mentally then their cones have a overlap or a similarity or shared space and but you can't really experience another party's cones or or can you indirectly um so that notion of knowing that really hooked me when i just came into the call that you have these cones going on at the same time these experiences that are through time and maybe when we read literature what we're doing is we're experiencing someone else's cone in the past in order to try to project it onto our experience in the future and then maybe you live by that rhetoric maybe the rhetoric actually becomes your future right so if your mind can have a certain state that gets acquired by contact with other states it makes a conformational change kind of like prionic governance we have a

conformational change in a protein when it misfolds maybe have a conformational change in your cone state when it comes in contact with another cone that has a different folding so that the narrative maybe there's a narrative element to the contiguousness of the cones and because they have a temporal and a physical element to them but you can only experience them in the present then the rest of it is just projection either in the past or the future and so maybe that's maybe the cone is entirely rhetorical and maybe it's convincing yourself and of the state that you have for existential continuity and the ways of knowing maybe goes to that in terms of multiple cone generators trying to get together and figure out if their little cones equal a big cone it's really just a fascinating way to visualize the space there's one other reference i want to make which is from the museum of jurassic technology in california culver city they have been displayed by a guy named uh theory by sono bend s-o-n-n-a-b-e-n-d and it's called theories uh theories of memory or theories of forgetting something like and his whole thing was cones reseeding and approaching light cones and the way he divides up space and knowledge is by taking different arcs and sections of these cones is really kind of fascinating stuff so really nice scott i'm reminded a lot um shanna dobson and like the way that she perceives time and like also in microscopy like you can take a projection of an image which is where like all of the stacks from the angles that you're using in the light cone come together and you can see it all at the same like point or it's like flattened out and it makes me think about visualizing time like that um daniel and then stephen so thanks scott i totally agree one of the big questions is about the intersections of time cones what kind of happens at that intersection do we think about them getting like folded up or having some new shape that arises and i think a way that we could have a simple hands-on intuition there is like with the two-dimensional game of life where there are certain shapes like gliders that kind of propagate on the landscape and then there's certain moving shapes perfectly annihilate each other there's certain ones that leave only rubble there are certain ones that replicate but then there's other ones that you know pass through each other depending on some feature interaction of the shape in the game and then there's ones that uh do computation like there's of course great videos of the game of life being set up to do an emulation of a of a computer like the ones that we're on but slower um so that really maybe connects a few pieces this sort of um agent-based interaction through time we get our intuition in this grid zone like the game of life and then maybe some of those interactions are happening in the natural systems just like some game theory mechanisms that were in the matrix of mathematics sort of also play out in biological systems really nice daniel uh stephen yeah this this question of time also relates to when we

say the now you know we know from even to say the participatory sense making that we're anticipating when something's going to happen often what we sense is is that there's a lag so in a way we're always a little bit behind what's just happened yet at the same time at certain times we're anticipating things before they happen and maybe before we realize we're doing the anticipating and then we've made the decision before we knew we made the decision and then we've done it so and which is where maybe reason partly why sports is so satisfying because we're we're making it have to be the baseball bat has to hit the ball and you're somehow bringing in that you know there's no you either hit it and it goes out there the baseball diamond or it doesn't but the same time when you made the the the cognitive process happened earlier the realization of what you actually hit it might be later than the now that you think you're in so there's this interesting window there of of what is now or can you even find the now um as i think shanna argued in in her math stream daniel so stephen with a sports example and that kind of relates us back to our uh the skilled performance discussions that we had previously so that reminds me of a bf skinner book beyond freedom and dignity where he's like if you imagine a situation where somebody is like unconscious or um they're forced to do something you don't give them that much agency or autonomy but then in the uh situations that look unprompted like somebody walking up on stage then playing like a violin performance which is still a mechanical performance we regard that highly and we shower that with the um sort of aura of agency so he was kind of questioning that from his own behaviorist perspective and it's just kind of cool to think about how active inference and multi-scale systems those same questions come up like there are cases where things seem really constrained as far as actions are concerned and then there's other times where behavior just seems like come out of nowhere from by like a bird song or something like that so having a framework for behavior that can connect the thoughts across those isolated occurrences is really promising nice and yeah we've talked about planning as inference and really the author brings up here um planning in relation to the homeostatic space right so so really this might be the first like like how planning came to be is like planning for some potential state that is not my ideal homeostatic space steven yeah both a good point yeah because if it's already my ideal homeostatic space why use all that energy valuable energy to try and predict it when it's already being done quite well and like daniel was saying there about skillful performance i mean once you get into this groove and someone's in the zone you know more and more of the processes have probably been taken care of by morphological computing which is basically the cellular processes and then when it's at these other temporal depths i i'm predicting and just

getting right the emphasis on the third note before the crescendo or whatever it is you know it may be that in some ways when you're in the flow which is that term when people are kind of immersed in an experience you know are you in some ways less conscious of more rapid dynamics and uh if without getting into a debate about consciousness i think we've got enough on the table but um but just that question about um yes it is it is encompassing many more areas of what we've talked about than that first had been apparent nice and this touches on memory also so i flipped over to that slide scott you know it's interesting that notion just a minute ago about being in an optimal or zone space um i heard somebody quote somebody a few years ago was talking about different personalities where some people like the social equivalent of classical music and some people like the social equivalent of jazz and so the people like the social equivalent of classical music want to hang out with the same people all the time and then oh bill how's bill's drinking problem going oh how's mary's you know uh kids doing you know it's the same story just incremental changes kind of like if you hear beethoven's fifth and it's like you're like whoa that's slightly off because your expectation is different whereas jazz you don't know what it's gonna be boop just all different things right so it's kind of interesting what your expectations are and what that optimization is jazz is a different kind of optimization right and so it made me think about the anterior cingulate nucleus my understanding is that's the part of the brain that if you're eating at a ripe berry bush and enjoying ripe berries the anterior cingulate nucleus if it fires off will make you think huh i wonder if there's a riper berry on the next bush and so it's the and and i i call it the commitment area where men can't commit to a relationship i believe it's probably the anterior cingulate nucleus they're looking for a riper berry so to speak so so that i wonder if there's an interesting rand i'll use a notion of random number generation here but it's probably not that but if there's a randomness factor that is in your optimum so is your state when it's optimized what is there something in the model itself that can throw randomness in there besides the externality so that and then you become like a novelty seeking person who has a highly active anterior cingulate nucleus is that an optimization which is superior for instance if you have a changing highly rapidly changing environment things like that just a couple of thoughts on optimization and what the zone might look like in different settings for different embodiments yeah that reminds me like i mean it's like a life plan or like an organismal life plan like some organisms like have to be in their niche environment like you can never take them out but then like some organisms like prepare for maximum randomness like i think about a tardigrade you know like you can do anything to them like you can dehydrate them and send them to the

moon and have them come back and all these things but but they are they're you know built they're well suited for maximum randomness daniel i liked scott's point there about the genre of optimization um i'm i'm personally always surprised listening to both jazz and classical it's just that it's a different genre and so a musician in either or both of those areas though would have like a deep enough generative model to know when one thing was off by even a tiny bit even in a song that might sound um discordant to somebody from because the improvisation tradition of one genre like might be more uh i don't know non-linear in some way but definitely this idea of like having interpretable or experienceable ways of interacting with different optimization processes different distributed processes are there more jazz like or more classical like uh collective behaviors and then also this broader question that is returned to centrally which is like that cognition is related to individuality that's really uh relevant for talking about where is cognition offloaded or how does cognition offload into the environment through development like through modification of the niche and then also through evolution where also the niche can be modified in the long term but evolution can select for like canalization so stabilization of things that were accessed plastically in one generation nice stephen and then scott yeah i mean falling in on what you're saying about the environment and and the jazz and classical um i used to live in prague for for a year or so and and what was interesting there so you've got a lot of classical music venues and operas and you've got lots of jazz as well but they're very different spaces as well right you've got the basement jazz bar in some cellar and you've got so there's a question as well about yeah once when you start going to one or the other you probably will keep going to one or the other for a while if that makes sense you sort of even though so beyond the music once you yeah there is a kind of um a dynamic because those those two are fairly clear it's relatively extremes i suppose even though they're both what you would call high art um in some ways as well so you've got that interesting thing about how you bring in the environment and that's one thing active inference does which many forms of psychology and that don't is how much is it about changing your generative model and how or how much is it about another model choosing your environment and letting the environment do a lot of the uh motivating for you um so that's interesting i was just also on a call um about metaphors and andrew austin who does that work he talks about if you want to motivate yourself it's actually easier to change your environment it's an externality that can be helpful if you want to change your choice making that can be internal so like don't try and choose get others to make better choices make better choices yourself you might not be able to motivate yourself maybe find things out there that help motivate so in the same way

that you know maybe that's good if i want to get a more ordered kind of structured view of the world for a while maybe i'll start going to some classical concerts for a while if i want to release and start sort of being more free-flowing maybe i'll go and watch some jazz um so anyway kind of interesting thanks scott i love all of you i gotta say this is i mean it's been a few weeks since i've been able to be on the calls just for different reasons and this is beautiful it's such a beautiful thing okay so so i was just showing you my tardigrade banjo ukulele which is what i was holding up to the screen so that's a target painted a couple of things so you know it's interesting love the last couple of comments and that notion of um access to other right and that kind of recreational and de-risking access to the other right you want to know the externality because you want to render it innocuous for survival purposes and then render it delicious for aesthetic purposes right and so a couple of books this is the the past is a foreign country by lowenthal which is kind of like okay temporally the assertion in the book is the past actually isn't a foreign country they're the same it's us right and so the same thing oh they're so different oh that's other right oh we've come beyond this but that's our own aspiration then another book introduction to comparative study of private law comparative law you got in law and language i my assertion that the mind exists in language and the brain is just an antenna tuned to it well law is a formalization of language and understanding so you have again comparative study there's another book i have i couldn't put my hands on it's the cambridge handbook of adaptation studies in theater and the articles on how do you adapt german theater for a greek audience or whatever right so all these things there's a exotic element to it right and which is both threatening and reaffirming and delicious and scary right and so these isn't it interesting that those states are we other than novelty perceiving either by resistance or seeking is is that the essence of consciousness and so if you had the equivalent of isothermy where you had it in isotherm you have temperature equivalence in two systems and you can't perform work because there's no temperature differential heat differentials well what if we had no meaning differential between two systems is that describable vis-a-vis that variable as one system and is that something that starts to get to you sociality or shared narrative and is that why that becomes so existential because the reinforcements of that becomes the amplifying wave essentially the amplifying embodiment becomes so powerful that it overtakes the system the system has no other internal state that's as reinforced and as amplified so therefore that one becomes overwhelming and maybe that's a source of why it's hard to crack misinformation by feeding fact or something like that anyway just a couple of random thoughts there really nice scott i um took the tony robbins test one time like what's

your driving force in life and i'm totally driven by instability which makes me like a desperate novelty seeking person um and i'm also reminded in your points about language i've been thinking a lot about language because of some projects i'm involved in and uh i was listening to josh tanenbaum speak yesterday about his um like uh program that he developed it's like a neural network neural network like a sleep wake helmholtz machine that actually like composes language i mean it doesn't it does some like phone numbers and it but it writes code like computer programs um and you know it draws symbols and it does these cool like things based on a framework that's um you know kind of analogous to active inference where the exploration phase happens in the sleep phase of the machine so exploration is like this recombination of all of the potential like factors right so like in the training phase it's it's trained on you know constructs the generative model or it constructs the recognition model based on all of the training input like if it's drawings it's going to be all of you know potential shapes and then it recombines these while it thinks about recombining them to kind of become a field expert it was really an interesting paper uh sorry to detour there daniel you go for it scott and then i'll give a thought uh thanks yeah just that's fantastic and it makes me think one of the things i've been fussing with for a while and you probably heard me say it is that the mind does not exist in the brain it exists in language and culture and the brain is tuned to the mind and the example i give is my niece who was adopted from china at five months old now has graduated from cornell she's looks like a chinese person she doesn't speak chinese she doesn't like chinese food she doesn't like chinese music she speaks spanish and english and she's a central pennsylvania gal right so her mind was not there's no feral consciousness and her mind was developed in central pennsylvania so what it makes me think what you were just saying is so interesting so maybe the mind where maybe our models are just so so much hubris there and we think the mind precedes communication and maybe it just doesn't maybe it is no mind affordance other than communication and language so like plato when he said the reason that authors were the challenge is you couldn't query the consciousness of the author maybe it was an even a deeper comment about mind that the queering of consciousness is mind itself and so maybe active inference is the function of mind it's not a mind seeking to to refine its state more but it is the fact of mind is the the act of active inference it just feels that was that was a shift that i hadn't thought about before and it gets away from centralizing mind individual brains that then come in contact to try to fix up their models and rather putting situating mind in the edge itself not the node from a graph theory perspective that's fascinating stuff uh daniel and then dave yep to kind of bring it to active inference but really just any kind

of dynamical modeling we shouldn't let this like snapshot govern the way that the system works because systems change through time so they're not just the snapshot picture maybe the prevalence of photography makes it easier sometimes to imagine this sort of snap shooting and active inference is hopefully providing a way to just think about it again whether instrumentally or more in this sort of like philosophical or metaphysical way which everyone is welcome to do but the actual uh core is just going to be leading to many questions for a long time dave one of the yeah one of the uh environments that is extraordinarily powerful at putting people in altered states of and especially distorting their time sense is a device an enter an entertainment device which was also a psychological research device called music color stephen may have seen references to that this was developed in the 1950s and was in some shows and clubs in london in the late 50s and early 60s um it's been mentioned a few times um andrew pickering mentions that a few other people in passing um and there was an exhibit that that was uh developed by the same engineer that tried to play in paris but got coveted out if none of you have seen that i should i have most of the material that's ever been captured as far as i can tell because it was a series of talks a series of literally chats uh um in the 1980s so i better gather that up and um it's all it's all in a single series on youtube that i have in a playlist so let's do that yeah the the guy would literally his musicians would believe that five minutes had passed when it was actually eight hours they had been playing solos for 10 hours at night risking a rest for trespass when they thought oh no no the guard you know everybody's still here i've only been here 10 minutes buying my flute and people would get so spaced out they drive their cars into the middle of roundabouts up on and bang into the posts because it just didn't occur to them not to do that or they'd they'd look into the the u tracks and watch the train come to them if people hadn't grabbed them and pulled them off what were you doing what do you mean i was i was looking at the lights and no drugs and they had fun and the bystanders had no idea this was happening because the oh these guys are just playing jazz but no they were on cloud9 that's cool it reminds me of a virtual reality daniel so it's interesting that with frameworks for communication like active inference we can think about syntactic repetition like repeating the same word semantic repetition and novelty like somebody saying the same thing differently but also narrative novelty and uh similarity or at least consistency coherency and so the other piece from what scott had said earlier was going back to some of our earliest streams on narrative and communication as active there's a ton of ways to think about that like is narrative like a summary statistic of the agent-based generative models like you got 25 at this parameter and 75 at this parameter and then narrative is just like a descriptive state but

what happens when narratives or various representations of them can actually through mechanisms feedback onto sensory and therefore internal so how do we think about narrative and active inference and narrative and communication as like higher order semantic structures that help reduce uncertainty allow for efficient information transfer all the imperatives of cell to cell communication bioelectric but it's happening on a spatial and a temporal scale that demands auditory transmission but of course can also be done electrically really nice scott this is i'm my again my face hurts from smiling so i so i had this book a rhetorical style it's called and it has dozens of rhetorical flourishes like the use of french suggests style and the use of german in words engineering excellence and then there's different placements i mean it's just all these different ways to construct the sentence all these differences and it's so fascinating that notion that is that what what's the cellular equivalent of rhetoric right is that when a cell wants to persuade let's say there's a growing tip of a mesenchyme tissue or some whatever in a plant or a human whatever and it's then and there's a decision to be made based on a turing wave that's coming in and what finger orientation you have right and turing wave responsible for finger growth there's a periodicity to the starting and stopping of the chemicals and that's why you have multiple fingers so what what's the notion of persuasion in non-human speech so a peacock's feathers are persuading the female of its fitness because the coloration one assertion is with coloration if it's a healthy looking color or the dance is healthily performed it suggests a healthy individual and that's persuasion for mating let's say i'm just asserting possible rhetoric and is all communication rhetorical in that it's made to persuade and the persuasion is not necessarily in it may be in the self-interest of both the speaker and the externality right because i'm talking about situations where you're running an internal model and you're trying to recruit other internal models which are your externality but for other people are there into or other things are there internal models right and so if you want to get some externality that's an organism to do something consistent with your model you need to er entrain them harmonically couple with them so is rhetoric what's the equivalent of rhetoric in other non-i guess non-language systems i use the word language in non-human language systems really nice um and i'm reminded of uh well it made me think of how does the narrative play across scales really like from the cellular to the organ level to the organism all stephen yeah i think that also bringing that into what daniel was saying was structure and narrative so there's um and actually that i just looking that up that color you know the color and how that wheel relates to sound music um so you've got these questions of when do structures start to be present that gives something else another type of geometry and where is

that present so it could literally be um there could be a type of geometric space in the way the neurons have arranged themselves and that could be a certain level at which they're getting fed in embodied information about light sources so it will only really happen when you've got an eye that's measuring light data that then starts to structure itself in that form and that could then underneath so and that could also apply to there may be a sort of a quality to the way sound is based on it being a whole level of you know hertz or half levels of hurts and half tones so there's something around i think there's an interesting point around when we have structure and we then this rhetorical piece may happen once we can oh crikey yeah so you got that as well the hell so you've got you've got you've got then you've got systems that we can see now the question with rhetorical stuff is is rhetorical stuff the stuff that we can see as a system that you can then argue your point you know some sort of way this and i suppose at another level the cells could come to a point where they could argue a case with each other and so there's some system in place that sits on that structure but the thing that always is in the background which is again what i was saying earlier most people doing systems don't think about when they talk about emergence they is is there some sort of swarming dynamics happening now whether this bioelectric stuff sits and adds a new layer i mean i noticed on one of the slides there it talked about kind of like a new layer of um uh like does this add a new layer to um to to this dynamic and i'm i'm curious about whether that your thoughts on that in in the sense of um you know do you do you see this as being an a new layer that's been added or is this somehow something which builds a bridge to the way the structures could emerge at the swarming um is it because it talks about memory i know yet the slide that talks about memory i can't quite find which one it is now but one of the slides talks about memory is significant in allowing the structuring to happen you know is it something about forming that memory in that structure or does the structure need to be there first i'm sort of putting that out to us and i don't know whether that's something we could ask michael but and maybe just refine that thought but it seems to be something which is a question that relates to how this all fits together nice dave yeah for scott um if you want to see how rhetoric is founded in the basic emotions and also get an biological evolutionary perspective on that read the chapter on play in yak poncep's affective neuroscience and once you've gotten through that look at the later piaget's later work on the role of play when he really got very subtle about the socialization and how how kids uh use play to become very expert in the in the ideal case in role taking and the creation of uh uh social uh relations and being able to play without anybody's feelings getting hurt and so on yeah the the concept of play i think is really

important um in the space of active inference because it really kind of represents that that exploratory component scott i'll give a thought while he stepped away it was related to his point about persuasion so if anything it's not going to be a binary it's going to be first off instrumental it's about how it's being modeled we're not looking for something that we're going to see under a microscope or it wouldn't even matter if we could we're talking about categorization so that we can reduce our uncertainty about action so the models that we set up could be a binary state model or not and that wouldn't be a metaphysical claim but just thinking about it as far as biological systems are concerned and taking it to that sort of like level like what if one molecule is secreted that sensitizes the cell to a second molecule you know is that type of sequencing would that be considered a minimal viable rhetoric or intracellular and uh just made me think about how there's sort of different types of interactions and that they convey different information something that my advisor professor gordon really explored a lot in the ant colony case how different types of interactions just update how likely an individual is to do different kinds of actions and that's seen across systems so sometimes that's just a simple enough statement like there's symbols that we see you know one or fifty or a thousand times a day and then there's things that are more explicitly narrated more rarely so then in the case of the human culture human narratives it just opens up the design space because there's multiple kinds of nice scott did you want to give a thought yeah a couple of things that is really nice daniel that's like kind of the notion in introducing something like discretion in a sense so if a system is running a let's let's call it active that running that model are they are they bringing their model to bear on the environment because it embodies the information of their past genetically and epigenetically and behaviorally so a three-legged dog is it running with three legs just because now it has three legs and that's it that's the affordance or or is there discretion in the active part of active inference that may be an interesting dividing point in terms of rhetoric is rhetoric can rhetoric be a persuasion brought by and thing that's intrinsic so a tiger jumps in front of me on a path and i'm walking the tiger is ferocious because it's an apex predator and has big teeth and i'm not is the tiger exercising discretion on being ferocious or is it just intrinsically ferocious to me right so is so that's that to me that feels like there may be something there on rhetoric if there's a choice maybe that the organism has or choices not maybe maybe the organism's not a single individual maybe it's a hive so that's one thing then that gets to that quorum sensing which is really rhetorical argument seems like it's because there there's a choice shall we swarm or not now when it's locusts and it's a pheromone for gregariousness right that's what switches the pheromone switches on and off

when they're in close contact so question is is that a discretionary thing when there's a pheromone impact you know again it doesn't have to be in either or it's just that maybe there's gradations of discretion that are exercised in different ways and so the other then it gets to me to get to synesthesia so if i could sense something um i could make a decision but instead i'm fed a pheromone that makes me feel make a decision without my conscious awareness is that still rhetoric and is that a quorum if it's a non-conscious quorum maybe it's a different state when it's non-conscious and also when you have internal synesthesia one of the things that's fascinating to me i once heard that the assertion that people with perfect pitch might be synesthetes because they're perceiving sound as color for instance and they can discern colors more readily in their mental capacity so therefore they can discern the notes better than people who can't discern notes as color so again all those things feel like they work together in terms of is the consciousness that's held in the mind held in the individual brain of an individual organism and or in the communications among the organisms and so maybe that takes the rhetoric question and renders it a nullity because maybe it's degrees of quorum that resemble discretion and maybe the internal acts of discretion are merely priors that were learned from an externality anyway anyway a few thoughts yeah i'm really reminded scott and this is going to turn you up a lot this is going to turn up your volume a lot the uh so i just learned about the work of um asifa majeed i'm not sure if i'm saying her name right but she does this amazing work on language how language shapes thought and how she thought shapes language um and she looks at at like in cultures they have you know there's was one i think it's the jahi culture they have like 17 different kinds of smells that they like smell and it's like you know one of them is like i mean we don't have words for smell we just have it smells like a rose or it smells like meat or whatever so she you know but they have actual in their vernacular they have different words for different smells and and like in english we have you know a gazillion words for color but like in some cultures the color blue and green are the same so so uh and you know it's the words for like one that was super amazing like we have all this word for like fingertip fingernail uh finger hand forearm bicep like a shoulder i mean all of this in some cultures is one word your entire arm every part of it like so you don't discern that there's no difference between your thumb and your shoulder like that's wild to me right and so how does that that shape the way that we think about it uh sorry if i'm rambling daniel no fun points so as far as how connectivity of communications related to rhetorical outcomes rhetorical ecosystems i think an analogy to look for would be in the connectivity of brain regions and of course in the spm toolkit and the dynamic causal modeling that we referenced a bit earlier

because there's brain regions that are anatomically connected so whether they're next-door cells or whether there's a long-range connection and then there's also like functional and effective connectivity which are different ways of looking at the signal of dynamical data and then drawing edges in a network where the nodes are the regions and then the edges reflect like a dynamical correlation or granger causation which does of course bring it into this whole dynamic causal modeling the whole question about multi-level cause cause across different uh parts of a system that aren't directly touching is it just our physicalism and our assumptions of materialism that like require something to be touching to have a direct influence is it just some sort of anti-you know non-locality type stance that's very deep prior is that something that when relaxed provides useful solutions or just ones that are a bit less susceptible to one argument but totally susceptible to other arguments nice questions very nice uh stephen and then dave yeah this idea like you the the question about structure and the relationship between how things can relate or touch i think it makes uh the way that we bring in date our sensorial information and these cones in a way with a cell you kind of each cell presumably can have this perception that can extend we have these multiple threads of perception that come in you know we've got the visual perception we've got the auditory perception we've got and then how do we structure our narratives well if we have a culture which structures the ability to smell think about smell in that sort of refined way compared to us who maybe gets very involved in the musculature and the nail and the you know the amount of time people spend now on their nails you know in in in our culture so that will change the structure of the narratives that you're going to tell each other and um that would be interesting and i actually just did a laban laban dance sort of workshop recently and they have they did the verticalities one of them like you've got the base which is the uh the feet and the floor you've got the bowl of the pelvis that you can sort of rock on you've got the diaphragm you've got the base of the mouth and you've got the top of the head as if they're almost different ways of grounding your ways of being right and as you go up and down i think that again speaks to that structuring how much would that change the structure of the narrative and that would change the structure of the memory and how does that structure of memory play into the structure and memory in paper you know could be interesting yeah the uh different ways of structuring language are quite striking to me my um tagalog is just dramatically different in the way it expresses ideas from the european languages there's no gender for instance um there are three cases nominative genitive and ablative and almost everything is expressed in all circumstances expects expressed in the ablative so where when for what purpose by means of what for whose

benefit that's all just expressed by putting however many of those causes there are into the ablative and if you need to be specific you throw particles in but they're always optional so uh if you challenge somebody well how do you know whether you're you know how what something means in in a sentence so well isn't it obvious if i talk about monday that when you translate it to english you'd say on and when we're saying when i'm saying at new york of course you would translate that to in and uh for well of course you translate it forth but when our before our son had been in the u.s for a long time he was constantly using this at for everything at three o'clock at manhattan at um pittsburgh at my girlfriend anything other than who's doing the action but who's doing the action isn't expressed in a straightforward way either so it's it's it's pretty wacky cool nice thoughts yeah sorry pretty wacky is what anybody with a highly precise language like russian or chinese would say you know when they hear somebody speaking even careful english this is wait a minute you had a not in your sense what is it exactly that you're negating there's no way to tell what an english speaker means once he's put not into a sense or in russian it's so strict and so specific same thing you know how do you know in english well you just figure it out what don't you just say what you mean no we don't let you figure that out definitely some ambiguity there i wonder like in cell to cell communication um if there's ambiguity like in in the bioelectric signal um like what which way does should a cell grow or go like touching back onto the the language of this multi-scale cognition daniel so i think to connect this uh language discussion to the um like the goal and the whole sensing point of the paper these different ways in which human languages are communicating state updates about the world people are able to live living people use all these languages so that says a lot it means they're able to be uptaken and transmitted by our cognitive apparatus and then also they're able to persist in a niche it doesn't mean that every language is going to be as successful in any niche but there is a place where it's successful so just like ants it's like that algorithms working might be working somewhere else but it might not be so then we can look at this amazing diversity of language and communication structures including ones that don't even only use audio or ones that use audio dimensions that we're not you know using certain sounds or something but then also the syntactic and then like what is the invariance what is actually being communicated that's the structure of what is is being shared between agents that are like one another in certain dimensions and they're sharing state updates so they can make it happen in their niche so it's like there's the total the diversity is because there's enough invariance for us to recognize it as part of the same category and then there's important things to look at in what it is that's in common yeah definitely um it's

definitely like interesting to think about how the the communication shapes the structure shapes the all of that the structure of society and the structure of a human body uh scott you know it's so you guys are this is so fantastic i this is like everything comes together in these conversations so i was i was thinking what is this reminding me of i just read an article and i was like is it this does it remind me of this article which is about universal chemical synthesis computing no was it about this highly multiplex spatial mapping of microbial communities sort of but no what this reminds me of is this space time from bits this guy i just read this article it's about quantum gravity and what he asserts is gravity is what we perceive as it's actually entanglement of bits of information and so if you take out the entanglement you have no space time and what i'm thinking is the entanglements we just described that's everything we call it mind we call it society we call it eusociality we call it space-time it's edges and we're characterizing these edges and our affordances are so constrained that we try to describe it but we're describing it it's the shadows in the cave right we're seeing in just indirectly what's actually out there and one of the things that's enriching about this active inference notion is it provides us with a framing that's sufficiently generic but also sufficiently robust that it explains things in the stack and the time stack and the space stack it's kind of like wow this may be how things interact maybe things are entangled and we use the word internal model and external reality but maybe the nature of rhetoric and maybe the nature of entanglement is the seeking of connection by neg entropy systems so i you know my life definition is um that living systems are auto catalytic and entropy secreting my son said i just wanted to use the word secreting because it sounded gushy but if they're auto catalytic and entropy secreting then if you're a system that's living you're going to want to seek out other living systems because you need reproductive and nutritional opportunities in order to be auto catalytic and and you know it goes to the frog nervous system rules right if you if frog encounters something that's smaller than it and it moves it eats it if it's larger than it and it moves it runs away and if it's the same size as it and it moves it mates with it very simple rules that let a frog be a frog right so it feels like maybe what we're banging up against is too much too many words that we're throwing at simple concept and maybe we're talking about is harmonic coupling of living systems that are seeking other neg entropy states either to consume or mate with the last point i have a poster that i got from a guy in san francisco and it's in my shop and it says futurism imagining a fake future in the hope that the real future will come along and mate with it and that's kind of like what we're we do in active inference we're imagining our way into a we're perceiving an externality just because our perception say okay there's

something you don't control out there and then we try to mate with it or eat it or run away from it i was reminded of the light cone slide here when we're talking about space and time and edges and um how that daniel included the slide like light is the boundary between space and time like that's kind of that trips me out um steven and then daniel yeah i think when we be interesting to also i think this opens up maybe a revisiting or with this with chris fields and some of this work quantum contextuality as well because as you start to because one thing he did mention in his presentation you know there's this question about when it goes from being contextual which is or quantum like which is a highly constrained set of options basically once you've got more options for movement things start to move into more classical even if there's loads of quantum stuff going on at the small scale they all average out and effectively the gas in my room is behaving similar to the gases in all the rooms that we're in a general generic sense only if we went into the sort of and and there's a question that comes up then around this sequence in um and um because where where is that sequencing possibly constrained enough that you might hit some things like that and is that part of what happens at these well you could call them meso levels i mean the cell the surface of a cell is small but it's not small from an atomic point of view it's still relatively aggregated yet we're getting these distinct flips charge and is it plausible that anything big as the brain can also do something and it may be that the term quantum is not necessary that in that sense but um yeah it'd be interesting to think about what it what what importance is there to be a constrained set of sequences a bit like daniel mentioned if a certain molecule is released and then another one's released and then another one's released that could be some sort of type of rhetorical dynamic but that there needs to be a constraint so that it's not just part of the classical noise you know it's not just swamped by a big flux of um randomness right that it actually is a sequence even if it's not as clear as you know so that might be an interesting question i don't know whether it's a fair question to put to mike if that's is or maybe it's more of a chris fields type question but i know they work together so they might have talked about some of definitely daniel and then scott yep it was a key theme in this paper and in cybernetics in general that like goal seeking and planning as inference is a big area that's why we're always talking about like control theory and it is interesting to wonder how does the bioelectricity play a role in goal setting and then in basically policy selection because you know active inference we don't just specify which trajectory is going to be like the preferred one and then condense it down to a single summary statistic it's like there's the preferred states with the c and then the affordances with the e and then there's a equation that is selecting policies based upon you know a combination

of different ways of phrasing some of the variables so then it's like how does that framing which is not just one way of going about doing what is it gonna allow us to say about like how systems actually plan i think that's kind of related to the hint in figure 2 on 32 like possible alien come ai it's saying that this framework wouldn't just help us describe systems but recognize systems that we didn't have ways of recognizing otherwise perhaps nice point and if um you know if another system was communicating with us how would we know right like what what would that even look like scott i was um have you guys seen this article decoding and perturbing decision states in real time now it's um p-e-i-x-o-t-o diogo pixoto it's it's about decision states it seems like it's very similar to what we're talking about here um 25th of march 2021 from nature magazine um and then bearheim v-e-r-h-e-i-n is another author um one of the things the reason the reason i was holding that up one of the things i was just writing down you know light cones lights um bosons right photons are bot or bosons and one of the things i always fuss with a little bit okay is there a difference between fermionic and bosonic physics here that we need to attend to so the idea of fermions i always think of them as firm so they make up matter and bosons make up energy stuff and information is carried by bosons but also carried by embodiments in in fermions and stuff and so it's so interesting to me that how does persuasion the rhetoric the the history get embodied both both sonically and fermionically again you have the peacock and you have the birds and the and the animals that develop certain physicalities over time as communication right it and it's a deep time communication and there's less discretion in it let's say they can exercise it or not but they have the embodi it's embodied in the physicality and then there's the action that they take which is again they have the physicality with which they act so what are the instruments given them physically but then they convey information by the behaviors they engage in and i keep coming back and thinking well it's kind of interesting if we look at the carrier signal again of information um is it in fact bosonic always it's things are carried by light i guess things carried by sound are being carried fermionically by the pressure in the sound wave i'm not sure so maybe the medium i was just wondering whether the medium relates to the notion of rhetoric and if there's a pure form of information transfer when it's bosonic rather than fermionic and maybe that's getting way too reductionist on the thing but it i wondered whether light cones in their purest form which presumably are involved light right the original light traveling out and it can only go this way and you can't be beyond it and blue when you said before it was a statement that what is it outside the light cone there is no it's like the defines the edge of space time for the observer and so what's what's out there is an externality it feels like that boundary when that's known to

other parties maybe in communication we're trying to do is think outside the light cone uh daniel and then stephen so the part of the light cone going forward in time is anticipation and if we take anticipation again not to be like a metaphysical claim about what systems are really doing but what the system is anticipating outwards that entails a generative model through of the states that matter for this kind of an agent which is the internal state of the model itself the blanket states which are sense in action expected outcomes and expected actions and then the state of the niche the external states so kind of operationalizing the light cone as a specific prediction on specific variables might be helpful and kind of give a little bit more insight into how it could be used in a computer and then also just to scout your point about like the media's the message essentially it's like a multi-scale media is the message it's not just um bold print type versus cursive it's actually like you know you're bringing up types of subatomic particles like that media you know is ether the ultimate media like the nature of the media does matter and yeah it's a trade-off space like with peptides you have a larger signaling space than with small molecules like the combinatorics are just different but that entails totally new challenges like with specificity or cross-activation of neuropeptides or how do you keep them linked through evolutionary time and then we see sort insights into those design trade-offs when we see like wow there are like a lot of neuropeptides that come from one precursor longer protein so that suggests that their co-regulation is held together through specific design principles so those are really interesting questions about how like the speed and the frequency and the bandwidth of different information channels structure how they end up performing yeah i think daniel you're making a good point and it feeds on with um this question about time scales is if we i assume that slides up the one that's there with the tick and the dog human and possible alien um and so the cones there yes it's very useful to think of it in terms of i've got my maximum information now maybe my maximum information was even a little bit in the past but it depends on what regime of attention i was applying right i i made choices so as i make choices into the present i reduce the possible cone extent i've got available you know um as i bring things only into my conscious awareness you know um certain things are going to be left out and then bringing that into okay this is a particular type of future i might predict based on my regime of attention is actually probably more useful in some ways to this because the cone going out it's a nice idea but it can lead to thinking that we know more and more in the future there's more more futures out there which is there is more more futures but not necessarily coming out from us um so it could be this is a bit more like okay my cone could move around a lot and there is actually a lot of analogy with that because now a lot of futures

work they're actually saying you're better off picking a scenario in the future building out the scenario in the future as a separate thing not inheriting your present because that will always bias you and then bring it back bring your present which is what you know and your assumptions here and sort of bridging the two in terms of the narrative rather than this idea of i can go because i'm could just be going with all my assumptions and all my pre-assumptions from the present and making all these stories about the and um and starting it almost starts from a singularity in those other diagrams it's like goes from a singularity and opens up whereas actually really the widest is now potentially this is the most information i have potentially available or at least within the 10 or 20 seconds of this moment um so i think that actually there's other ways that this could change the way people think about sense making in time and space really super nice point stephen um scott wow stephen wow i mean that's so awesome when it makes me think okay first of all encoding and decoding okay so something i i just wrote down mutations encode new speculative space time so mutation gives you access it's a random mutation that gives you access to space time you weren't intending to access and the same thing with externalities if another thing comes in then this is jazz this is jazz genetics right is you get mutations and it's like whoa i didn't want to access that that new like cone area but i guess i'm there now okay let's see what we got here and that's pretty awesome because otherwise you'd be settled in your ways and you'd never see that space that this flight cone area would never open up and you don't know what it's going to be you don't know it's like it's i always said to my kids life is a one-act play no rehearsals that's a light cone right you just go through and it's like okay here we go so that idea of encoding and decoding is kind of interesting so you have this the we we go out there in the world and we're like okay i want to interpret the world now some things were encoded with intention to communicate music speech some things were encoded with intention to observe right cryptography stuff whatever some things were encoded with no intention to communicate but they do communicate like oh leaves are green so therefore over time we figure out that they have magnesium in the photosynthetic cycle or whatever right so they're not the color is encoded in the thing but the tree is not intending to say i'm doing photosynthesis right it's just we now take that perception and can decode something that was never encoded with discretion i guess or intention by the organism or whatever you know we don't run going with that so this notion of these light cones and it from the perceiver's perspective we can decode within a light cone we can decide an artist decodes a light cone and somebody else seeing the same perceptions would not have the same light cone right somebody with different priors can see things a

person who knows history will not repeat it because they're like they see the same history now and they go whoa whoa whoa whoa that looks like an insurrection like their light cone is more trained to the possibility that's an insurrection so this last couple of comments are fascinating to me that we're moving through space we have this active inference model we're running a model there's myriad other models and non-models out there that we're seeking to decode some subset of them were intended to be coding information some subset or not and we're just moving through space decoding everything some of it will be false signal some of the actual signal and so the light cones the experience we have obviously can never be replicated by another person with different priors but also in real time it's a i don't know what the equivalent is of a cacophony of light cones a babble of light cones or it's like that ice cream cone again there's too much ice cream maybe in the cone anyway it's fascinating notion of that over decoding in a way and what and maybe the scientific method has something to do with a rigor on the over decoding that can happen with alchemy or something like that nice point daniel and then stephen so to scott's point about the mutation as uh accessing spatial temporal uh zones like expanding uh into space-time i thought okay maybe there's some like quantum you know 4d three plus one interpretation of that how can we think about that with the information that we already empirically know and it's actually that mutations in developmental biology developmental genetics facilitate the exploration of new developmental space and time like by slowing down or speeding up the way that a certain cell type is uh susceptible to a hormone or something like that so what's being played with is actually the level faster because that's why there's the intergenerational transmission of those mutations but then what they're actually modifying like the source code but not exactly in that way of course is how the system bootstraps and modifies itself through development and learning and then how can we think about active inference in that way well we can think about the mutations as modifying like the initial parameters or some of the state variables for the parameters that are important for an active inference agent like the structure of its internal generative model its blanket states like its sensory and action affordances and then the niche so that's a way where we can have the generality to talk about like potentially a system across different scales but then when we're ready to actually formalize it and then ask how to design communication in that niche nice point stephen yeah that's a good point thinking about those different time scales and and sort of following on from that you've got this question of the um so maybe the things are happening at this light speed and there's a lot of electrical activity happening in these cells and groups of cells and and we've got these structures that we can come together and of course we

then have this intermodal integration between different cells nerves coming in and the question then also is as it's coming in say from my body to my brain and my feet are seeing the world in the past cons compared to my nose right because it takes longer for the sort of information to travel up from my foot the signals at the speed of light traveling but by the time it does that through all the chemical interactions it you know it's milliseconds not billionths of a second so it's coming together in this structure of the body which is grown to you know to integrate it but at some point the system has to calibrate it has to calibrate so that it doesn't think that my foot is in a different time to my nose or my ears or my fingers right so that sort of brings that question of um you know some of this multi-scale stuff will come out of these kind of electrical light speed dynamics but then it's going to also come into the uh the structures i think the daniel was talking about structures we all sort of mentioned structures and there's going to be a system that sort of at times has to sort of tweak it you know and maybe that system's not needed when you've got a small set of cells that are all firing electrically and doing other chemical signals they just the structure's enough in a way but uh um so yeah i thought that that might be kind of that might be an interesting question to ask michael as well if this you know how does how does that kind of play out with this nice daniel and then scott yeah just a quick comment on that so the signal processing or the sort of outside in perspective would suggest that there would have to be like a recalibration then the generative model out way of thinking is that the generative model is that like your hands clap and the sound and the visual are the same event so there isn't as much of a calibration of inputs as there is a reconciliation of deep temporal models which are timed based upon the inferred causal structure of the world which is usually not two separate events but rather one and then that is actually embodied in the sensory relay which does take different amounts of time to transfer from different parts of the body nice scott yeah okay so i just was just one thing on that i'm just before scott that's a good point i think that's interesting you know thinking about predictive processing and not getting caught up in the in that's where it's tricky because it's it's not directly an input it's a kind error message so yeah that's cool that's not the part that's experienced yeah and the parsing of that opens up awareness right if you can parse the pieces and start to break it down this last couple of pieces i thought you know the occupy movement occupy wall street it feels like active inference we should have t-shirts made up occupy light cone because everyone's every organism is trying to occupy everyone else's light cone i think i mean and when i so we worked for microsoft for at the law school we had a program microsoft paid professors intellectual property professors to go to china to teach intellectual

property professors in china and in india usip law concepts now why why was microsoft doing that they were doing that because they were trying to render the externality innocuous there was a piracy externality that they were dealing with and so they thought if they teach the teachers in those jurisdictions about the value of intellectual property and then teach the lawyers at the law schools the teachers teach the lawyers those notions then the microsoft notions of intellectual property they'll be left software privacy so that was they weren't doing it for beneficence they were doing it to render the externality innocuous they were trying to occupy the light of a bit of a culture right with this affordance uh shifting by mutating the teachers views and the way they taught intellectual property law so that that to me there's a i once heard a quote that trees were bullies because trees suck up the water and the light in an area right you know they bully other plants and i always thought you know trees are so nice and i always think of trees as bullies now and it's kind of interesting when some when an organism is applying rhetoric or other vehicles to sweep in and you know like pigeons when they do pigeon racing when you try to pigeons try to flock and you try to grab everyone else's pigeons right or i don't know in hive behaviors if hives pick up other bees and that makes them fitter as hives or if there's mixing of bees when they do swarming i don't know the answer to that i just don't know enough about bee biology but it is kind of interesting that light cones maybe are pretty darn aggressive for living things trying to render externalities innocuous and maybe that goes to gaia theory that ultimately what what's the light cone for gaia yeah the biggest light cone of them all just what you were saying made me think of maybe you know we think about it as a light cone but maybe what we're looking at is like an information cone you know and so like a concentration of information within an individual because in the end we dissipate you know there's no more information concentrated there uh stephen yeah and it might be a question when we think about this term light cone and i was walking home last night and i could see the clouds and i could see the lightning but i could only hear the thunder you know five seconds late or whatever so the information cone on the audio was different to the information if it is an information cone and then i don't know the smell of the the rain or the change of that so um i might be interested in how much the there may be some elements in certain scales where really it's light it's about the speed of light i mean near enough so that's what it is or maybe it's a broader category which could roll out into these other spatial temporal modalities yeah the uh so i mean you know when you're in a space if you wake up in a dark room what's the first thing you do like where am i right turn on the light that's a very human um way to maximize your information gain i mean not every organism obviously can

see like uh daniel i think my closing thought on this would just be that it really speaks steven's story about how there actually was a temporal structuring even though you cognitively knew it was a single event you actually classified it into a separate kind of event which is like the exception that proves the rule that suggested long long ago that the speed of sound and light were different how are we going to make sense of this total continuum of signaling in the niche we can think about ostensive cues and helping you plan there's something when you smell something and you have eight hours to plan for the storm versus you know you hear the lightning strike you have a half second to plan et cetera so we can think about the structuring of different kinds of cues in the niche as giving information that in light of a generative model able to be used for anticipation in action so that's just speaking to how core active inferences to thinking about these different processes coming and that's why it'll be awesome to get the author's perspective on all of this and what active inference is going to do nice scott you have a final thought for us yeah just again that i'm still intrigued by the notion of the entanglement creating space time and this whole discussion of time cones and the time cones in the relation of space-time relativity theory etc and uh i feel like one thing that active inference has done whatever it's and there's so it's powerful in many ways one thing it's done is allowed me to open up my mind to processes that feel like they're replicated across time and scales and that's interesting because whether or not they're strictly fall within active inference math and rigor the the model has uh opened up so many different ways of thinking about things and the entanglement notion where we think of two is it two particles or one spooky action at a distance it feels like everything else we're talking about active inference is spooky action at a distance for me it's just that it's a different spooky action at a distance and so it's fascinating the notion of entanglements and it feels like light may be describing the things that are amenable to in be entangled or coupled and so maybe if it's an information cone that's an notion that we can start to say communities expand out that which is amenable to coupling because they widen the light cones so that using that may start to be a way that we can illustrate the benefits of community-based behaviors and things like that so we can start to bring this back and say look you can't see around the corner but that person who's up there they're standing there and they can see around the corner and if they can talk to you they can tell you what's around the corner so maybe that's good right and so maybe that's contact tracing for pandemics maybe that's you know giving the neighborhood watch for other toxic things in the neighborhood environment whatever so it feels like this gets simple enough you know people may not want to know about light cones or whatever but they it starts to get the

entanglement and that notion of cause and effect feels like there's a story that we can tell here and illustrate pretty simply anyway it's exciting it's an exciting conversation some wonderful stuff and it's delightful to be back in the conversation so just uh to come to the author's like final point in this um you know the dissolution of the self right this is like the the practice of of zen um he says it is striking that the process which zen practice is meant to reverse attachment to past memories and high valence for future expectations or fears is precisely the process suggested to be responsible for the creation of complex selves so i want to know what happens to the informational light cone in this like dissolution of the self does it like you know turn into lots of little light cones as is kind of implied in like the the one figure but or does it just like poof and like dissipate i don't know it's it's interesting to think about um that for me he says it's unclear whether it's beneficial or even possible to truly live in the moment and let go of past memories and future expectations but anyone who succeeded in doing this would achieve precisely what zen promises the dissolution of the self um so thanks everyone this has been a super interesting discussion and i am looking forward to our conversation hopefully with the author next week thanks nice work blue thanks thanks very much take care have a great day bye everyone